

1. Low Contrast Image

| A grayscale image appears dull with very little difference between object and background intensities.

(a) Interpretation of the scenario and cause of low contrast

A low-contrast image has pixel intensities that are all clustered within a narrow band of the available grey-level range (e.g. 256 levels for an 8-bit image), instead of being spread across the full range. Because the object and background intensities fall close together in this narrow band, they appear visually similar, making the image look dull and flat. This typically happens due to poor lighting conditions during capture (too little or too even illumination), incorrect camera exposure settings, limited sensor dynamic range, or atmospheric/optical scattering that compresses the intensity range.

(b) Key issues affecting image visibility

- The histogram is narrow and concentrated in a small range, rather than spread across 0-255.
- Object and background pixel values are too close together, so edges and boundaries are hard to distinguish.
- Fine details in both shadows and highlights are compressed and effectively lost.
- The image is visually flat, reducing usefulness for human interpretation or for downstream processing such as segmentation or edge detection.

(c) Applying an appropriate enhancement technique

Contrast stretching (a piecewise-linear grey level transformation) is applied. If the pixel values in the image lie within a narrow range $[r_{min}, r_{max}]$, each pixel is remapped to the full range $[0, 255]$ using:

$$s = ((r - r_{min}) / (r_{max} - r_{min})) \times 255$$

This linearly expands the narrow band of intensities so it spans the entire available range. If the image's histogram is more irregular or non-linear, histogram equalization can be used instead, which redistributes grey levels so the histogram becomes more uniform, automatically boosting contrast in the most densely populated intensity regions.

(d) Justification

Contrast stretching is chosen because it directly targets the root cause of the problem - a narrow intensity range - by expanding it to use the full dynamic range, without altering the relative order of pixel intensities (so no false detail is introduced). It is simple, computationally inexpensive, and preserves the structural relationships between object and background. Histogram equalization is a reasonable alternative when the useful intensity range is not clearly known in advance or the distribution is uneven, since it adapts automatically to the image's histogram shape.

(e) Summary

The dull appearance is caused by a compressed intensity range; the key issue is insufficient separation between object and background grey levels; and contrast stretching (or histogram equalization) resolves this by remapping the narrow range to the full grey-level scale, restoring clear visual distinction between objects and background.